METRIC
A-A-55806
March 25, 1996
Superseding
MIL-B-81744A(1)
December 23, 1976

COMMERCIAL ITEM DESCRIPTION

BARRIER COATING SOLUTION, LUBRICANT MIGRATION DETERRING

The General Services Administration has authorized the use of this Commercial Item Description, for all federal agencies.

1. Scope. This description covers a lubricant-migration-deterring barrier coating solution. This solution is intended for application to selected surfaces of precision ball bearings to form a thin oil-repellent film which retains the lubricant within the bearing and reduces lubrication migration.

2. Salient characteristics.

2.1 Materials. The barrier coating solution shall be composed of a 0.18 percent solution of poly 1H, 1H-pentadecafluorooctyl methacrylate in a fluorochemical solvent.

2.2 Abrasion resistance. The barrier coating film shall be capable of retaining a silicone lubricant from migrating across or through the barrier film for a period of 24 hours in a vertical plane after a one stroke abrasion of 0.94 to 1.4 Kg (2 to 3 lbs.) force against the barrier film.

2.3 Contact angle. The barrier coating film shall be capable of maintaining a contact angle of not less than 60 degrees with a drop of n-hexadecane formed on the barrier film.

2.4 Thermal stability. The barrier coating film shall be capable of retaining a silicone lubricant from migrating outside of the barrier film during and after exposure to a temperature of 175° ± 2°C (347° ± 3.6°F) for 24 hours.

2.5 Nonvolatile content. The barrier coating solution shall have a nonvolatile content of not less than 0.18 ± .005 grams.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VCA, 8000 Jefferson Davis Highway, Richmond, Va. 23297-5610.

AMSC-N/A FSC - 6850

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2.6 Contamination in solution. As an observation for solution contamination after being placed in unit size containers, a distilled water drop shall spread and not form a bead when placed in a sample of barrier film which has been exposed to a reagent grade acetone until the acetone has evaporated.

3. Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. Quality assurance provisions.

4.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of the Commercial Item Description, and that the product conforms to the producer's own drawings, specification, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for shipment shall be in accordance with American Society for Testing and Materials (ASTM) D 3951 or as specified in the contract or order.

6. Notes.

6.1 Addresses for obtaining reference documents.

ASTM
American Society for Testing and Materials
1916 Race Street
Philadelphia PA 19103

6.2 Ordering data.

a) Title, number, and date of this description.
b) Specify special packaging requirements.
c) Contracting officer may request proof of certification of commercial item prior to first contract delivery.

6.3 Known sources of supply.
3M Company
St. Paul MN 55101

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MILITARY INTEREST:
Custodians

Navy - AS
Army - AR
Air Force - 11

Review Activities
Army - MI, AV
Navy - OS, SH
Air Force - 99
DLA-GS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity: AS

Agent: DLA-GS

Project Number: 6850-1148

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